

Placement Empowerment Program
Cloud Computing and DevOps Centre

**Create a Simple Backup Script Create a script that
backups up your entire Git repository to a local folder
daily**

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Introduction and Overview

In this Proof of Concept (PoC), we will learn how to host a static website on an AWS EC2 instance. This process involves setting up a virtual machine in the cloud, configuring a web server, and deploying a simple static webpage. By following these steps, you will gain hands-on experience in cloud-based hosting, an essential skill for deploying and managing web applications.

Objective

The goal of this project is to:

1. Launch an AWS EC2 instance running Ubuntu.
2. Install and configure a web server (Apache or Nginx).
3. Deploy a simple static website.
4. Ensure the website is accessible via a public URL.
5. Understand basic security configurations for cloud-based hosting.

Importance of Cloud-Based Hosting

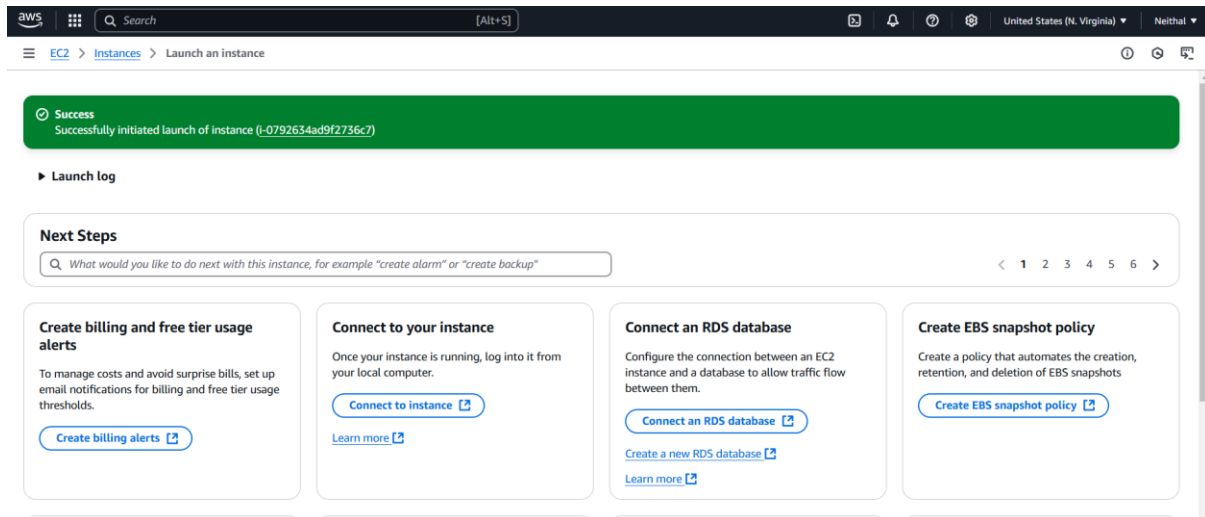
Hosting a website on a cloud-based virtual machine offers several advantages:

- **Scalability:** Easily upgrade resources as traffic grows.
- **Cost-Effective:** Pay for only what you use with cloud providers' flexible pricing.
- **Reliability:** Cloud infrastructure ensures high availability and uptime.
- **Security:** Advanced security configurations protect the website from vulnerabilities.

Steps for Execution

Step 1: Launch an AWS EC2 Instance First, log in to the AWS Management Console and navigate to the EC2 dashboard. Choose to launch a new instance and select Ubuntu as the operating system. Select an appropriate instance type, such as t2.micro, which is eligible for the AWS Free Tier. Configure security settings to allow HTTP and SSH access, then launch the instance and download the private key (.pem) file.

Step 2: Connect to the Instance Once the instance is running, copy its public IP address. Open a terminal and use SSH to connect to the server using the private key file. Ensure that the key file has the correct permissions before initiating the connection. Upon successful connection, you will have remote access to the cloud-based virtual machine.

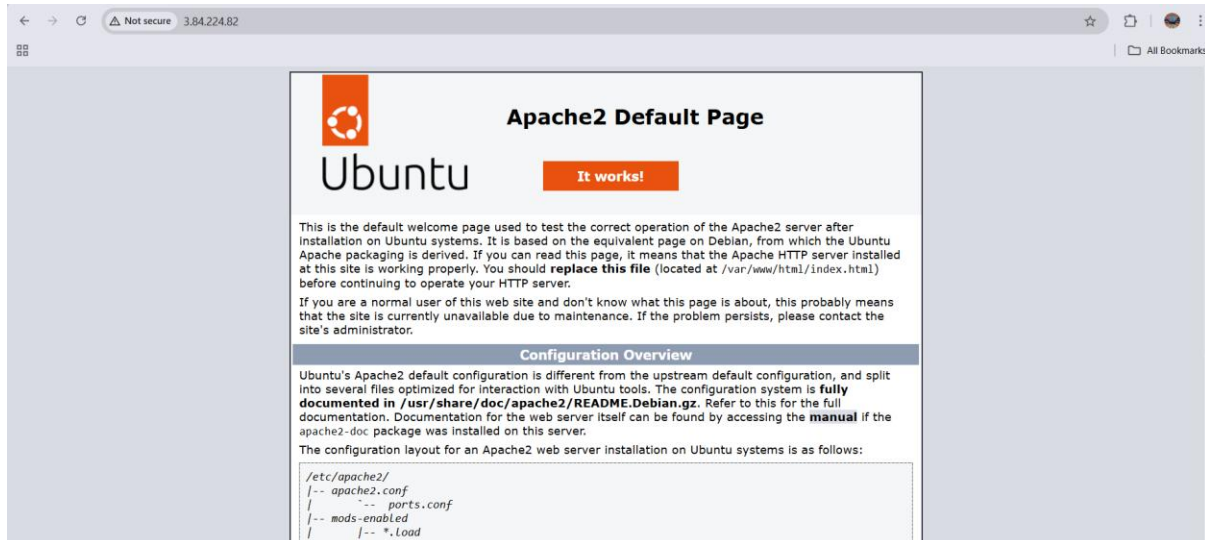


Step 3: Install and Configure a Web Server After connecting to the EC2 instance, update the package lists to ensure you have the latest software. Then, install the chosen web server, such as Apache or Nginx. Start the web server service and configure it to run automatically on system boot.

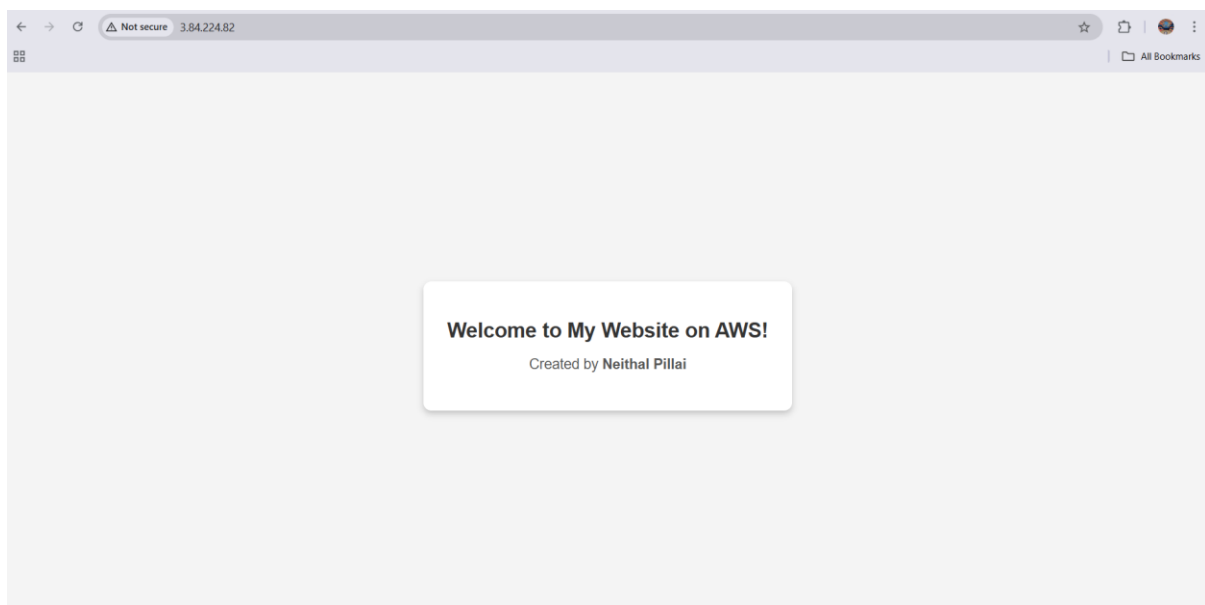
```
ubuntu@ip-172-31-87-198: ~
See "man sudo_root" for details.

ubuntu@ip-172-31-87-198:~$ chmod 400 my-key.pem
chmod: cannot access 'my-key.pem': No such file or directory
ubuntu@ip-172-31-87-198:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.1 LTS
Release:        24.04
Codename:       noble
ubuntu@ip-172-31-87-198:~$ free -h
               total        used        free     shared  buff/cache   available
Mem:           957Mi       310Mi       422Mi       888Ki       376Mi       647Mi
Swap:            0B           0B           0B
ubuntu@ip-172-31-87-198:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  1.7G  5.1G  26% /
tmpfs            479M   0    479M   0% /dev/shm
tmpfs            192M  876K  191M   1% /run
tmpfs             5.0M   0    5.0M   0% /run/lock
/dev/xvda16      881M   76M  744M  10% /boot
/dev/xvda15      105M   6.1M   99M   6% /boot/efi
tmpfs             96M  12K   96M   1% /run/user/1000
ubuntu@ip-172-31-87-198:~$
```

Step 4: Deploy the Static Website Navigate to the web server's document root directory and replace the default webpage with the static website content. Ensure that the permissions are correctly set so that the server can access and display the website files. Restart the web server to apply the changes.



Step 5: Verify and Access the Website Obtain the public IP address of the EC2 instance and enter it into a web browser. If the configuration is correct, the static website should be displayed. If necessary, adjust security group settings in AWS to allow public HTTP traffic.



Conclusion

By completing this PoC, we successfully deployed a static website on an AWS EC2 instance using a web server. This process introduced key concepts in cloud-

based hosting, including virtual machine setup, server configuration, and security considerations.

What I Learned from This Process

- How to launch and connect to an AWS EC2 instance.
- Installing and configuring a web server on a cloud-based VM.
- Deploying and managing static web content.
- Understanding security settings for cloud hosting.
- Troubleshooting common issues in web server setup.

This PoC serves as a foundational step towards hosting and managing web applications in the cloud effectively.